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Multiarterial Grafting and Survival After Coronary Artery Bypass Grafting

An Instrumental Variable Analysis

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ABSTRACT

BACKGROUND Randomized trials evaluating multiarterial grafting (MAG) vs single arterial grafting (SAG) during coronary artery bypass grafting (CABG) have not demonstrated a long-term survival benefit, whereas conventional retrospective studies have consistently reported improved survival with MAG. Whether this discordance reflects true treatment effect heterogeneity or bias from unmeasured confounding in observational analysis remains unclear.

OBJECTIVES Our objective was to evaluate whether the apparent survival advantage associated with MAG in conventional observational analyses persists after accounting for unmeasured confounding using a quasi-experimental instrumental variable (IV) approach and to assess the implications of these findings for long-term survival in an older Medicare population.

METHODS We retrospectively analyzed Medicare beneficiaries who underwent CABG from 2001 to 2019. Surgeon MAG rate during the 12 months preceding each operation was leveraged as an IV. Flexible parametric survival models with time-dependent effects were developed with MAG vs SAG as the exposure variable. The non-IV model adjusted for patient demographics, pre-existing comorbidities, hospital and surgeon characteristics, and procedural details. The IV model incorporated these same covariates plus the IV (surgeon MAG rate) using a 2-stage residual inclusion approach. Regression standardization was used to derive standardized survival probabilities and their differences.

RESULTS Among 1,291,314 beneficiaries, 1,145,760 (88.7%) underwent SAG and 145,554 (12.3%) underwent MAG. In the non-IV model, MAG recipients had improved risk-adjusted median survival as compared with SAG recipients: 10.74 years (95% CI: 10.70-10.79 years) vs 10.33 years (95% CI: 10.31-10.35 years), a difference of 0.41 years. Across 4,164 surgeons, the MAG rate during the 12 months preceding the index CABG was 7.7% $\pm$ 9.5% in SAG recipients and 32.9% $\pm$ 25.8% in MAG recipients. In the IV model, MAG recipients had similar risk-adjusted median survival compared with SAG recipients: 10.38 years (95% CI: 10.29-10.48 years) vs 10.38 years (95% CI: 10.35-10.40 years), a difference of 0.01 years.

CONCLUSIONS MAG was associated with a modest improvement in long-term survival in a conventional risk-adjusted analysis. However, this association was not robust to a quasi-experimental analysis in which surgeon MAG rate was incorporated as an IV to address unmeasured confounding. The contrast between these models suggests that traditional observational studies may overestimate the survival benefit of MAG because of unmeasured or difficult-to-measure patient characteristics that influence a surgeon's decision to offer MAG. (JACC. 2026;■:■–■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

2SLS = 2-stage least square

2SRI = 2-stage residual inclusion

ADI = Area Deprivation Index

BIMA = bilateral internal mammary artery

CABG = coronary artery bypass grafting

CMS = Centers of Medicare & Medicaid Services

CPT = Current Procedural Terminology

ICD = International Classification of Diseases

IV = instrumental variable

LIMA = left internal mammary artery

LRT = likelihood-ratio test

MAG = multiarterial grafting

MBSF = Master Beneficiary Summary File

SAG = single arterial grafting

SIMA = single internal mammary artery

STS = Society of Thoracic Surgeons

Coronary artery bypass grafting (CABG) is the recommended treatment to improve survival and reduce major adverse cardiovascular events in patients with complex coronary artery disease.[1,2] The importance of the left internal mammary artery (LIMA) as a bypass conduit was established by Loop et al, who demonstrated its superior patency rate compared with saphenous vein grafts and its association with improved survival and freedom from cardiac events.[3,4] After publication of these landmark reports, LIMA use during CABG increased from 30% in 1988 to 95% by 2010.[5,6] LIMA as a single arterial graft (SAG) with supplemental vein graft bypass conduits remains the dominant revascularization strategy in the United States, comprising >85% of CABGs.[7-10]

The success of the LIMA has fueled interest in additional arterial conduit use.[11] Randomized studies have demonstrated favorable arterial graft patency rates, and observational studies suggest that multiarterial grafting (MAG) during CABG is associated with improved survival.[12-16] A recent retrospective study from the Society of Thoracic Surgeons (STS) database, the largest to date, found that MAG was associated with superior survival compared with SAG.[17] However, the only randomized controlled trial adequately powered to evaluate long-term clinical outcomes of MAG found no difference in 10-year survival between patients randomized to the use of the bilateral internal mammary artery (BIMA) vs a single internal mammary artery (SIMA).[18] Thus, currently available retrospective data and randomized evidence on MAG are contradictory.

Conventional retrospective analyses comparing MAG and SAG are subject to confounding from both measurable and unmeasurable factors that may influence a surgeon's conduit selection during CABG.[19] Variation in surgeon MAG utilization can be leveraged as an instrumental variable (IV), introducing quasi-random variation in treatment assignment and allowing assessment of the extent to which unmeasured confounding may bias conventional observational studies.[20,21] Accordingly, the primary objective of this study was to examine whether the discordance between conventional observational studies and randomized evidence can be explained by unmeasured confounding. To this end, we compared long-term survival among Medicare beneficiaries undergoing isolated CABG with MAG or SAG using both conventional risk-adjusted analyses and IV-based models. Although this approach does not definitively adjudicate the comparative effectiveness of MAG vs SAG for all patients, it provides clinically relevant insight into long-term survival among older adults undergoing CABG when unmeasured confounding is more rigorously addressed.

METHODS

STUDY DESIGN. Administrative claims from the Centers of Medicare & Medicaid Services (CMS) between 2001 and 2019 were retrospectively reviewed. Records before 2001 were excluded, as coding data distinguishing on-pump from off-pump CABG was introduced that year, and we planned to include this variable in our risk-adjustment models.[22] The Baylor Scott & White Research Institute Institutional Review Board approved this study.

STUDY POPULATION. A 100% sample of Medicare beneficiary hospitalizations involving CABG was identified from the MedPAR file using International Classification of Diseases (ICD) procedural codes (Supplemental Table 1).[23] Beneficiaries without continuous Medicare Part B coverage or those enrolled in Medicare Advantage plans within 1 year before the index procedure were excluded, as such records may incompletely capture pre-existing comorbidities.[24] Records missing 9-digit zip code data or lacking a corresponding measure of Area Deprivation Index (ADI) were also excluded.[25]

Surgeon-billed Current Procedural Terminology (CPT) codes from the Carrier (Fee-for-Service) file were matched to each hospital admission, as previously described.[26] Records missing CPT codes describing operative details or lacking surgeon identification associated with CPT claims were excluded to minimize procedural mischaracterization. To ensure a cohort of isolated CABG procedures, we used a doubly adjudicated approach in which ICD (Supplemental Table 1) and CPT (Supplemental Table 2) codes were required to concordantly indicate that CABG was performed and that no concomitant cardiovascular procedure (Supplemental Tables 3 and 4) was present (other than surgical ablation of atrial fibrillation and/or coronary endarterectomy. In addition, cases involving redo mediastinal access, all venous grafting, or single-vessel CABG were excluded. Finally, to ensure an adequate characterization of surgeon preference for SAG vs MAG, beneficiaries treated by surgeons performing fewer than 10 CABG annually among Medicare beneficiaries were excluded.

Given the stability of CPT coding across the ICD-9/10 transition, CPT codes were used to define operative characteristics (Supplemental Table 2).[26] CABG was classified as MAG if CPT codes indicated use of ≥2 arterial conduits, regardless of conduit type, and SAG if CPT codes indicated use of a single arterial conduit; total graft counts (both arterial and venous grafts) were tabulated. This approach was validated in a multihospital analysis in which CPT-based conduit classification across 5 hospitals demonstrated high concordance with operative reports.[27] Because this validation was conducted within a single health system, national variation in coding practices and residual exposure misclassification cannot be excluded; such misclassification would not be addressed by IV methods, which rely on accurate treatment classification.

PATIENT DEMOGRAPHICS, COMORBIDITIES, AND SOCIAL DETERMINANTS OF HEALTH. Beneficiary demographics were ascertained from the Master Beneficiary Summary File (MBSF). Pre-existing comorbidities were identified using the Chronic Conditions segment of the MBSF and confirmed as pre-existing if the earliest documented date of diagnosis preceded the date of surgery.[24]

Race and ethnicity were ascertained using the CMS imputation algorithm,[28] which has been shown to improve the sensitivity in identifying minority race and ethnicity based on third-party survey data.[29] Neighborhood deprivation was assessed using the ADI metric, a composite measure of key social determinants of health that influence cardiovascular outcomes.[25] National ADI values from 2015 and 2019 were used: 2015 values for beneficiaries undergoing surgery from 2001 to 2015 and 2019 values for those undergoing surgery from 2016 to 2019. Sensitivity analyses demonstrated a high correlation between 2015 and 2019 ADI values, supporting an assumption of temporal stability for this metric.[30]

HOSPITAL AND SURGEON VOLUME. Annual CABG volumes were determined using the hospital CMS certification number linked to ICD codes for CABG and the surgeon's National Provider Identifier or Unique Physician Identification Number associated with CPT codes for CABG. Medicare CABG annual volume served as a surrogate for total CABG annual volume, a marker of quality at both surgeon and hospital levels.[31]

STUDY ENDPOINT. The primary endpoint was all-cause mortality. Vital status and date of death were determined using the internal Medicare documentation of death contained in the MBSF.

DEFINING THE IV: SURGEON MAG FREQUENCY. Surgeon practice patterns, specifically surgeon MAG frequency, were used as an IV to account for unmeasured confounders influencing conduit selection during CABG.[32] This approach assumes that variation in MAG use reflects surgeon preference rather than patient characteristics, thereby generating quasi-random variation in treatment assignment based on how patients are distributed among surgeons.[20,21] For validity, IV analyses rely on 3 key assumptions: relevance, in that surgeons differ meaningfully in their use of SAG vs MAG; exclusion, in that patient assignment to surgeons is unrelated to the surgeon's MAG preference; and independence, in that a surgeon's use of MAG is unrelated to other treatment decisions that directly affect outcomes (eg, surgeons with higher MAG rates have similar operative volumes and comparable rates of coronary endarterectomy).[32,33] We defined the IV as the surgeon's MAG rate during isolated CABG over the 12 months preceding the index operation. This IV was calculated dynamically as a rolling measure: For each index surgery, all isolated CABGs performed by that surgeon within the prior 12-month window were identified and the proportion performed with MAG was calculated, allowing the IV to capture contemporaneous surgeon preference at the time of each operation. Alternative IV definitions were evaluated in sensitivity analyses.

STATISTICAL ANALYSIS. Annual trends in MAG and SAG use during isolated CABG were assessed using the Cochran-Armitage test. Survival was compared by observed treatment type before and after risk adjustment. Risk adjustment was performed by including covariates (patient demographics, pre-existing comorbid conditions, and hospital, surgeon, and operative characteristics) in a flexible parametric survival model (non-IV model) parameterized on the log-cumulative hazard scale.[34] Restricted cubic splines (using default knot placement) were used for age, neighborhood deprivation (ADI), hospital volume, surgeon volume, and baseline hazard function to account for nonlinearity. Spline terms were iteratively tested with increasing degrees of freedom (up to 5) in a factorial fashion, and the final specification was selected according to the Akaike and Bayesian Information Criteria. Time-dependent effects for the exposure (MAG vs SAG) were included to relax the proportional hazards assumption, allowing the treatment-outcome association to vary over follow-up time. These time-dependent effects were modeled using restricted cubic splines with 3 degrees of freedom for the interaction between treatment and follow-up time

Then, in a second model (IV model), we applied a 2-stage residual inclusion (2SRI) IV approach to account for potential unmeasured confounding of the exposure-outcome association.[35] This control-function-based method has been shown to reduce bias in the settings of time-to-event outcomes and is recommended for IV analyses with nonlinear second-stage models.[36] The strength of the IV was assessed using either the likelihood-ratio test (LRT) or F-statistics, as appropriate.[33]

Regression standardization while fixing the treatment variable (MAG vs SAG) was applied to both the non-IV and IV models to obtain standardized (marginal) survival probability and difference curves (Supplemental Methods).[37] Survival probability predictions are known to be more robust to model misspecifications and to the choice of additive vs multiplicative hazards scales in IV settings.[38] Differences in survival probabilities have the natural interpretation as marginal risk differences, avoiding the interpretational challenges associated with HRs.[37,39] Our implementation of the 2SRI approach with flexible parametric survival models was benchmarked and validated against the available implementation in the *ivtools* R package,[40] which could not be used for this work because it does not implement standardized survival probability predictions.

To assess the robustness of our findings, we conducted an extensive set of sensitivity analyses. Each analysis was performed in 2 distinct cohorts: a restricted cohort of beneficiaries with ≥10 years of potential follow-up (CABG performed 2001-2009) in which 10-year survival could be evaluated as a dichotomous outcome, and the full cohort (CABG performed 2001-2019), allowing all beneficiaries to contribute to time-to-event survival analyses. This dual-cohort structure ensured that results were consistent when evaluated in beneficiaries with complete long-term follow-up (required for models using 2-stage least square [2SLS] estimation) as well as in the full contemporary sample.[33]

Using the 2 outcome specifications above, we first performed a logistic regression using the 2SRI approach.[35] We then applied a linear IV model using 2SLS estimation.[33] Next, we conducted time-to-event analyses, beginning with a Cox regression model estimated using the 2SRI method.[36] Finally, we implemented flexible parametric survival models using 2SRI, first assuming proportional hazards and then incorporating time-dependent effects.[37] This final specification is presented as our primary analysis.

Additionally, we evaluated alternative IV definitions, including a surgeon's MAG rate during the 36 or 60 months preceding the index operation as well as a surgeon's overall MAG frequency from 1999 to 2019. For comparison, we also report estimates from identical models that excluded the IV variables.

Data processing and statistical analyses were conducted using SAS version 9.4 (SAS Institute) and Stata software, version Stata/MP 19.5 (StataCorp), respectively. All estimates are presented with 95% CIs in brackets.

RESULTS

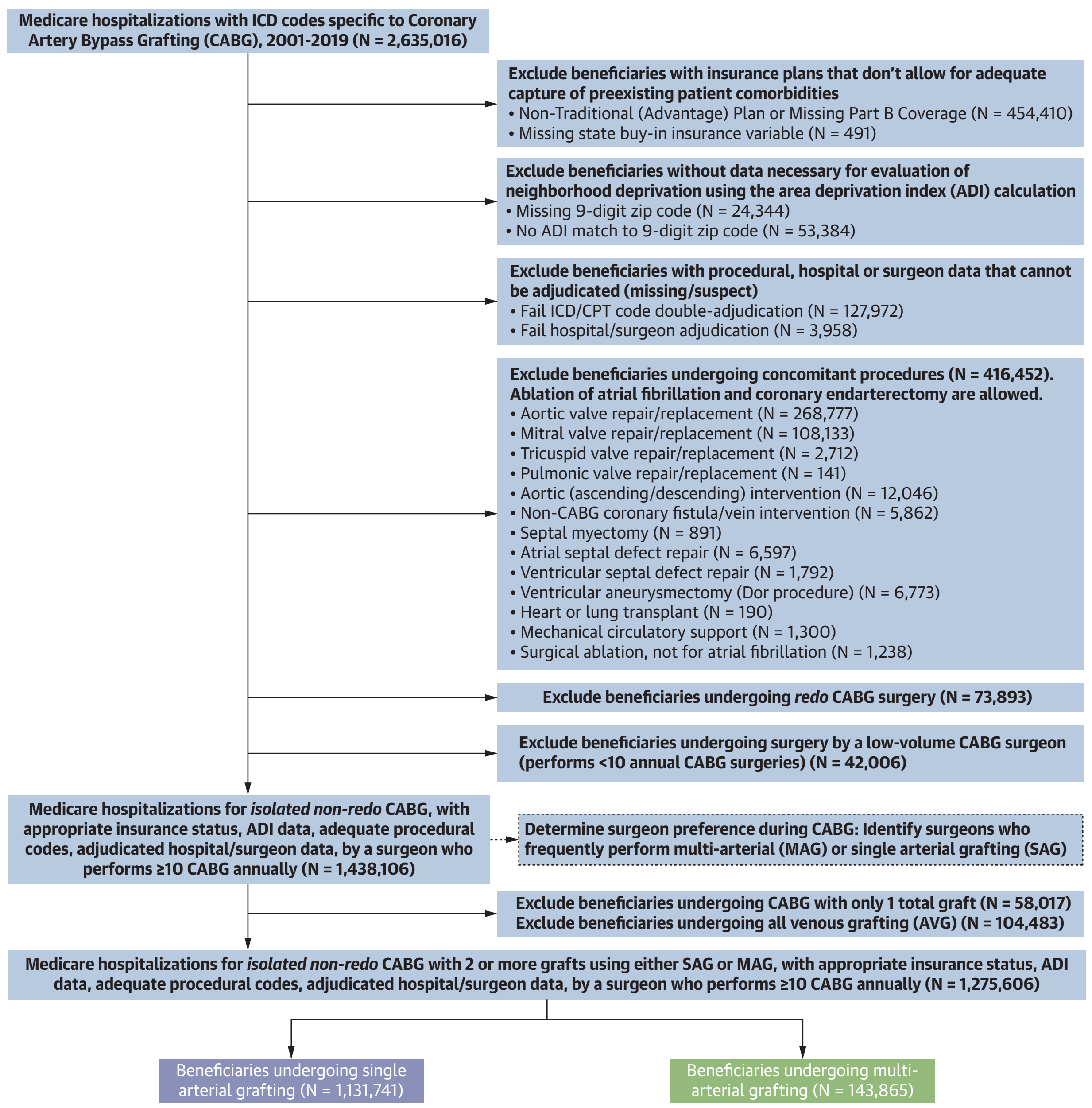
PATIENT POPULATION AND MAG UTILIZATION. We identified 1,438,106 Medicare beneficiaries who underwent nonredo, isolated CABG performed by surgeons performing ≥10 annual Medicare CABG cases from 2001 to 2019; this cohort was used to determine surgeon grafting frequencies (**Figure 1**). Among these patients, 58,017 who underwent single-vessel CABG and 104,483 who underwent all venous grafting were excluded. The final study cohort consisted of 1,275,606 beneficiaries, of whom 143,865 (11.3%) received MAG and 1,131,741 (88.7%) received SAG. Median follow-up time was 12.22 years (95% CI: 12.20-12.24 years).

The annual prevalence of SAG increased from 73.8% in 2001 to 87.3% in 2019 (Cochran-Armitage Z-statistic: 142; $P < 0.001$). MAG use declined from 11.9% in 2001 to 7.9% in 2015 (Cochran-Armitage Z-statistic: -52; $P < 0.001$) and then rebounded to 10.1% by 2019 (Cochran-Armitage Z-statistic: 9; $P < 0.001$) (**Figure 2**).

Demographics, comorbidities, and procedural characteristics for beneficiaries undergoing MAG and SAG are summarized in **Table 1**. Compared with MAG recipients, SAG recipients were older, more often women and of minority race, more likely to reside in the southern United States, more frequently lived in neighborhoods of higher deprivation, and had higher rates of pre-existing comorbidities. MAG recipients were more likely to undergo off-pump CABG, and MAG was more commonly performed earlier in the study period. The mean total number of grafts (arterial plus venous) was 3.8 ± 1.1 for MAG recipients and 3.4 ± 0.9 for SAG recipients. Supplemental Figure 1 shows unadjusted survival for MAG and SAG recipients.

RISK-ADJUSTED TREATMENT TYPE ANALYSIS. The non-IV flexible parametric survival model including time-dependent effects was generated using available covariates (LRT: 6.13; $P = 0.10$ comparing model assuming proportional hazards with model

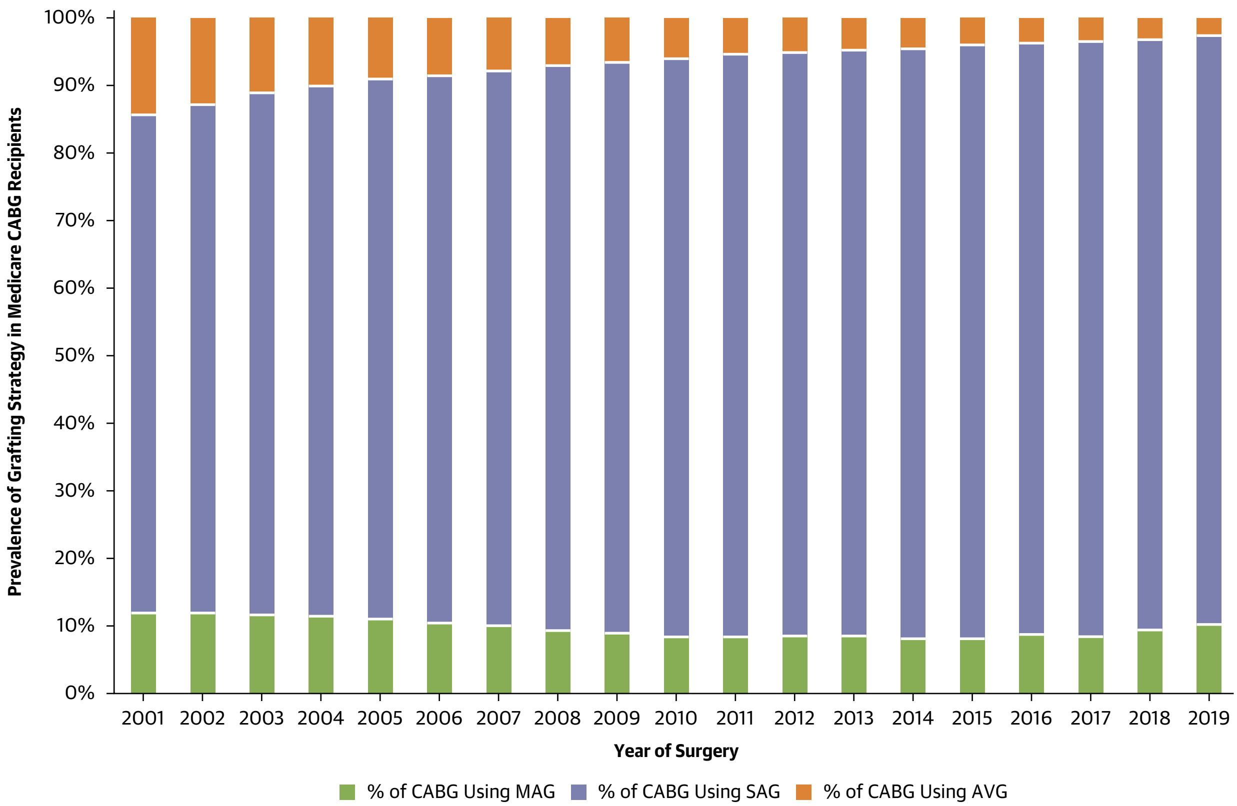
FIGURE 1 **Cohort Selection Diagram**



CABG = coronary artery bypass grafting; CPT = Current Procedural Terminology; ICD = International Classification of Diseases.

including time-dependent effects) (Supplemental Table 5). From the non-IV model, risk-adjusted standardized survival probabilities at 30 days, 1 year, 10 years, and 19 years after MAG vs SAG, respectively, were 98.0% (95% CI: 97.9%-98.0%) vs 97.9% (95% CI: 97.8%-97.9%), 92.5% (95% CI: 92.4%-92.7%) vs 92.0% (95% CI: 91.9%-92.1%), 53.8% (95% CI: 53.4%-54.0%) vs 51.7% (95% CI: 51.6%-51.8%), and 16.4% (95% CI: 16.1%-16.6%) vs 14.9% (95% CI: 14.8%-15.0%) (**Figure 3**). Risk-adjusted median survival was 10.74 years (95% CI: 10.70-10.79 years) and 10.33 years (95% CI: 10.31-10.35 years) for MAG and SAG recipients, a difference of 0.41 years.

FIGURE 2 **CABG Conduit Strategy Over Time**



Prevalence of conduit strategy (AVG, SAG, or MAG use) over time in Medicare beneficiaries undergoing isolated CABG. AVG = all venous grafting; MAG = multiarterial grafting; SAG = single arterial grafting; other abbreviations as in Figure 1.

INSTRUMENTAL VARIABLE. We identified 4,164 surgeons who performed ≥10 Medicare CABGs annually. For each surgeon, the rates of all venous grafting, SAG, and MAG among all isolated CABG cases performed during the 12 months preceding each index operation were calculated using data from 2000 to 2019 (Figure 4). Patient demographics, preexisting comorbidities, and operative characteristics stratified by quartile of surgeon MAG rate are provided for reference (Supplemental Table 6). In contrast to the substantial imbalances observed between MAG and SAG recipients (Table 1), patient clinical characteristics were generally well balanced across quartiles of surgeon MAG rate. However, variables related to surgeon and hospital characteristics (annual volumes, off-pump utilization, year of surgery) and geographic factors (Health and Human Services region, ADI) demonstrated standardized mean differences ranging from 0.15 to 0.37 (above the usual 0.10 threshold for balance), reflecting the institutional and regional clustering of surgeon practice patterns. These measured covariates were directly adjusted for in the second-stage outcome model of our 2SRI approach; residual imbalances would only bias IV estimates if they reflect imbalances in correlated unmeasured variables.

The LRT statistic comparing models with and without the IV was 176,733 ($P < 0.001$). The corresponding F-statistic from the 2SLS IV regression model was 368,158, indicating a strong instrument (Table 2).

RISK-ADJUSTED IV ANALYSIS. A flexible parametric survival model incorporating surgeon MAG rate during the prior 12 months as an IV (2SRI method) including time-dependent effects was developed using available covariates (Supplemental Table 7). Model fit was superior compared with the model assuming proportional hazards (LRT: 19.5; $P < 0.001$). From the IV model, risk-adjusted standardized survival probabilities at 30 days, 1 year, 10 years, and 19 years after MAG vs SAG, respectively, were 97.9% (95% CI: 97.8%-98.0%) vs 97.9% (95% CI: 97.8%-97.9%), 92.4% (95% CI: 91.9%-92.7%) vs 92.0% (95% CI: 91.9%-92.1%), 52.0% (95% CI: 51.5%-52.4%) vs 51.9% (95% CI: 51.8%-52.0%), and 15.4% (95% CI: 15.0%-15.9%) vs 15.0% (95% CI: 14.9%-15.2%) (**Figure 5**). Risk-adjusted median survival was 10.38 years (95% CI: 10.29-10.48 years) and 10.38 years (95% CI: 10.35-10.40 years) for MAG and SAG recipients, respectively, a difference of 0.01 years. The risk-adjusted standardized survival differences between MAG and SAG recipients in the non-IV model and IV model are depicted in **Figure 6**.

SENSITIVITY ANALYSES. The results of repeated sensitivity analyses using alternative IV approaches, IV definitions, and both the entire cohort and the restricted cohort with ≥10 years of follow-up are summarized in **Table 2**. Across all non-IV analyses, MAG was associated with improved survival. In contrast, every IV model leveraging surgeon MAG rate as an IV yielded a null association between MAG and long-term survival.

DISCUSSION

In this national analysis of Medicare beneficiaries undergoing isolated CABG, we applied both conventional observational and IV analytic approaches to evaluate the association between MAG and long-term survival. Our findings demonstrate a clear divergence between these methods: although traditional risk-adjusted analyses suggested a modest survival advantage associated with MAG, this association was not observed when surgeon MAG use was incorporated as an IV to address unmeasured confounding. These results do not resolve the open questions related to the clinical effectiveness of MAG vs SAG for all patients, and we caution readers not to interpret our findings as such. Rather, our findings suggests that the survival benefit of MAG consistently reported in conventional retrospective studies may be driven, at least in part, by unmeasured confounding related to patient selection and surgeon decision-making. Although this study primarily contributes a methodological perspective on why observational and randomized evidence in this field have diverged, it also provides clinically relevant insight into long-term survival among older adults undergoing CABG when unmeasured confounding is more rigorously addressed.

TABLE 1 Demographics, Comorbidities, and Procedural Characteristics of the Study Population, Stratified by MAG vs SAG

	CABG-SAG (n = 1,131,741)	CABG-MAG (n = 143,865)
Age	72.0 ± 7.8	70.8 ± 7.8
<65 y	119,410 (10.6)	17,402 (12.1)
65-69 y	287,187 (25.4)	43,231 (30.1)
70-74 y	293,555 (25.9)	37,515 (26.1)
75-79 y	249,689 (22.1)	28,656 (19.9)
80-84 y	140,628 (12.4)	13,481 (9.4)
≥85 y	41,272 (3.7)	3,580 (2.5)
Male	784,232 (69.3)	107,223 (74.5)
Race or ethnicity		
White	966,058 (85.4)	126,918 (88.2)
Black	68,321 (6.0)	6,605 (4.6)
Hispanic	61,742 (5.5)	5,910 (4.1)
Asian American, Native Hawaiian, and Pacific Islander	17,787 (1.6)	2,048 (1.4)
Health and Human Services region		
1 (HQ: Boston)	50,336 (4.5)	6,434 (4.5)
2 (HQ: New York)	87,505 (7.7)	14,168 (9.9)
3 (HQ: Philadelphia)	114,831 (10.2)	17,100 (11.9)
4 (HQ: Atlanta)	293,019 (25.9)	27,215 (18.9)
5 (HQ: Chicago)	217,264 (19.2)	30,184 (21.0)
6 (HQ: Dallas)	158,558 (14.0)	12,913 (9.0)
7 (HQ: Kansas)	65,385 (5.8)	12,080 (8.4)
8 (HQ: Denver)	23,872 (2.1)	6,346 (4.4)
9 (HQ: San Francisco)	89,831 (7.9)	10,332 (7.2)
10 (HQ: Seattle)	30,916 (2.7)	7,050 (4.9)
Area Deprivation Index	55.1 ± 26.8	50.7 ± 26.9
Insurance: state assists with Medicare premiums	152,115 (13.4)	15,808 (11.0)
Dialysis dependence	41,494 (3.7)	2,936 (2.0)
Chronic Conditions Warehouse comorbidities[a]		
Hypertension	961,481 (85.0)	117,070 (81.4)
Hyperlipidemia	877,226 (77.5)	108,593 (75.5)
Diabetes	527,679 (46.6)	60,309 (41.9)
Ischemic heart disease	105,273 (97.7)	140,892 (97.9)
Prior acute myocardial infarction	171,396 (15.1)	19,844 (13.8)
Prior congestive heart failure episode	393,649 (34.8)	42,600 (29.6)
Atrial fibrillation	122,306 (10.8)	13,768 (9.6)
Chronic kidney disease	269,859 (23.8)	26,452 (18.4)
Chronic obstructive pulmonary disease	278,085 (24.6)	28,674 (19.9)
Prior hip fracture	12,583 (1.1)	1,200 (0.8)
Depression	217,470 (19.2)	24,871 (17.3)
Stroke or transient ischemic attack	153,178 (13.5)	16,069 (11.2)
Dementia and/or Alzheimer disease	54,496 (4.8)	5,133 (3.6)
History of cancer	133,371 (11.8)	15,535 (10.8)
Admission urgency		
Elective	537,634 (47.5)	73,938 (51.4)
Urgent	294,708 (26.0)	36,694 (25.5)
Emergent	295,968 (26.2)	32,713 (22.7)
Year of surgery	2008 (2004-2013)	2006 (2003-2012)
Hospital annual CABG volume	175 (105-297)	190 (117-298)
Surgeon annual CABG volume	57 (36-84)	59 (37-86)

Continued on the next page

TABLE 1 Continued

	CABG-SAG (n = 1,131,741)	CABG-MAG (n = 143,865)
Surgeon rate of conduit strategy[b]		
All venous grafting rate, %	7.7 ± 9.5	5.1 ± 7.4
SAG rate, %	84.6 ± 14.7	61.9 ± 24.7
MAG rate, %	7.4 ± 7.4	32.9 ± 25.8
Concomitant surgical ablation of atrial fibrillation	26,935 (2.4)	3,175 (2.2)
Concomitant coronary endarterectomy	24,538 (2.2)	2,924 (2.0)
Off-pump CABG	188,265 (16.6)	32,599 (22.7)
Total grafts	3.41 ± 0.93	3.80 ± 1.11
1	0 (0.0)	0 (0.0)
2	178,179 (15.7)	18,880 (13.1)
3	465,420 (41.1)	38,559 (26.8)
4	358,418 (31.7)	49,682 (34.5)
5	107,740 (9.5)	25,586 (17.8)
≥6	21,984 (1.9)	11,158 (7.8)
Arterial grafts	1.00 ± 0.00	2.30 ± 0.58
0	0 (0.0)	0 (0.0)
1	1,131,741 (100)	0 (0.0)
≥2	0 (0.0)	143,865 (100)

Values are mean ± SD, n (%), or median (IQR). Standardized mean difference between columns. Health and Human Services regions pertain to the location of the patient residence at time of surgery, stratified into 10 Health and Human Services regions. [a]Pre-existing comorbidity data (comorbidities present before the date of surgery) were obtained from the Chronic Conditions segment of the Master Beneficiary Summary File. [b]Surgeon rate of conduit selection was calculated using conduit strategies used during all isolated CABG in Medicare patients performed by that surgeon during the 12 months preceding the index surgery. For example, all venous grafting rate = no. of all venous grafting cases / (no. of all venous grafting cases + no. of SAG cases + no. of MAG cases).

CABG = coronary artery bypass grafting; HQ = headquarters; MAG = multiarterial grafting; SAG = single arterial grafting.

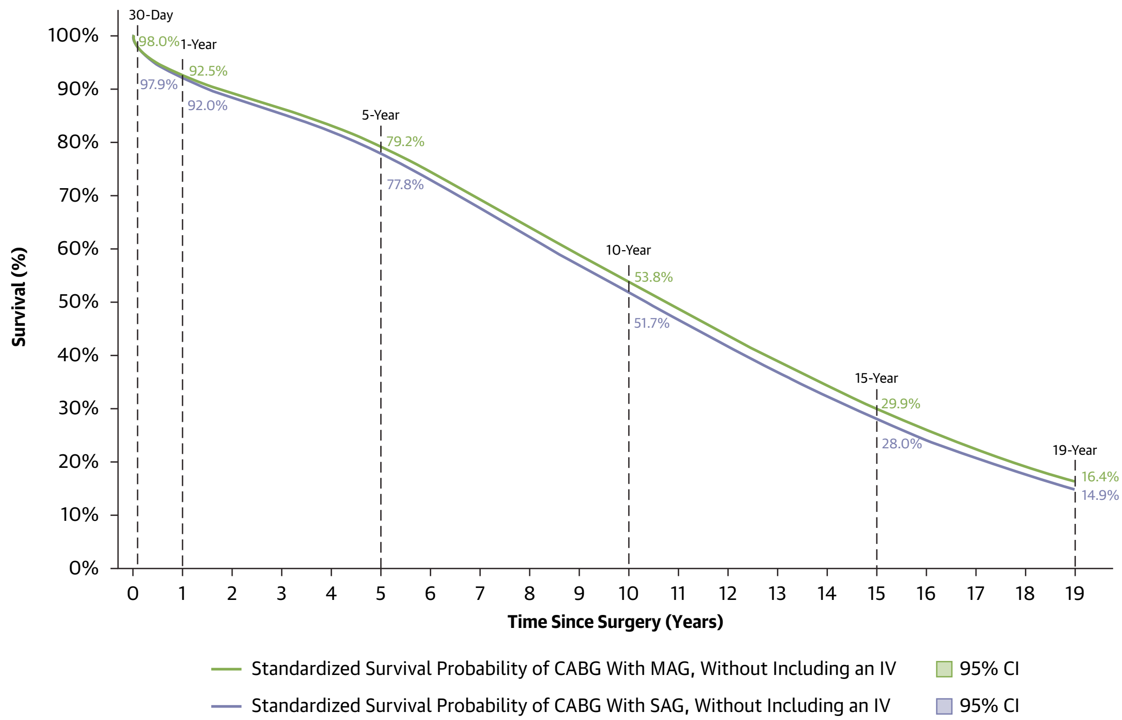
These findings underscore the limitations of conventional retrospective studies of MAG and reinforce the continued importance of prospective randomized controlled trials to determine whether MAG confers a meaningful survival advantage. Retrospective studies associate MAG with improved long-term survival, yet these findings are not consistent with randomized data, including the ART trial,[18] which to date have not demonstrated a survival advantage for MAG. This discrepancy may reflect unmeasured confounding in observational analyses, whereby factors not captured in retrospective datasets influence both a surgeon's conduit selection and a patient's long-term survival after CABG.[19] In this study, we evaluated the association between MAG (compared with SAG) and long-term survival using conventional risk-adjusted and IV approaches, enabling direct comparison of these methods and providing insight into the extent to which unmeasured confounding may influence conventional analyses.

In the conventional risk-adjusted analysis, MAG was associated with a statistically significant but clinically modest improvement in long-term survival of 0.41 years (approximately 5 months). These results are consistent with numerous prior retrospective observational studies, many of which have used propensity matching to mitigate confounding.[41] For example, the PRIORITY group reported improved 10-year survival for patients undergoing CABG with BIMA vs SIMA grafting after propensity matching 11,000 patients.[42] A meta-analysis including 79,000 patients from 27 observational studies also demonstrated increased long-term survival in patients who had BIMA compared with SIMA grafting.[43] Most notably, a recent analysis of the STS database matched 100,000 MAG recipients with 100,000 SAG recipients and demonstrated a survival advantage associated with MAG at 10 years.[17] Importantly, the absolute difference in 10-year survival between MAG and SAG recipients as measured in our analysis (53.8% vs 51.7%, difference of 2.1%) was similar to the difference reported in the STS analysis at 9 years (81.2% vs 78.8%, difference of 2.4%). Although the lower survival observed in our cohort likely reflects the older Medicare population compared with the STS population (mean age: 72 vs 65 years), the similarity in absolute survival differences across these 2 studies strengthens confidence in our conventional risk-adjusted findings and highlights that this modest association was attenuated when IV models were applied.

Indeed, valid concerns have been raised regarding whether unmeasured confounding variables influence both conduit selection and long-term survival, thereby biasing retrospective propensity-matched comparisons.[19] In clinical practice, factors such as coronary target quality, severity of upstream stenosis, conduit availability, patient frailty, and other poorly measured predictors of life expectancy (ie, patients who appear higher risk based on surgeon "eyeball" assessment) directly impact conduit selection and are also likely to influence long-term survival. These variables are difficult to capture in administrative or even detailed clinical databases and thus cannot be directly accounted for through standard analytical approaches to risk adjustment.

When surgeon MAG use was incorporated as an IV to address unmeasured confounding, median survival differed by only 0.01 years between MAG and SAG recipients, and standardized survival probabilities were nearly identical across follow-up. Under the previously described IV assumptions, analyzing outcomes based on surgeon MAG use enabled quasi-random variation in treatment assignment, potentially balancing unmeasured confounding variables across treatment groups.[20,21] Together, our findings indicate that the clinically modest survival association observed in the conventional risk-adjusted analyses was not robust to analytic approaches designed

FIGURE 3 **Long-Term Survival After CABG (Non-IV Model)**



Standardized survival probabilities after CABG in the non-IV model (a model *not* including surgeon MAG rate as an instrumental variable); stratified by whether the beneficiary received SAG (purple) or MAG (green). IV = instrumental variable; other abbreviations as in Figures 1 and 2.

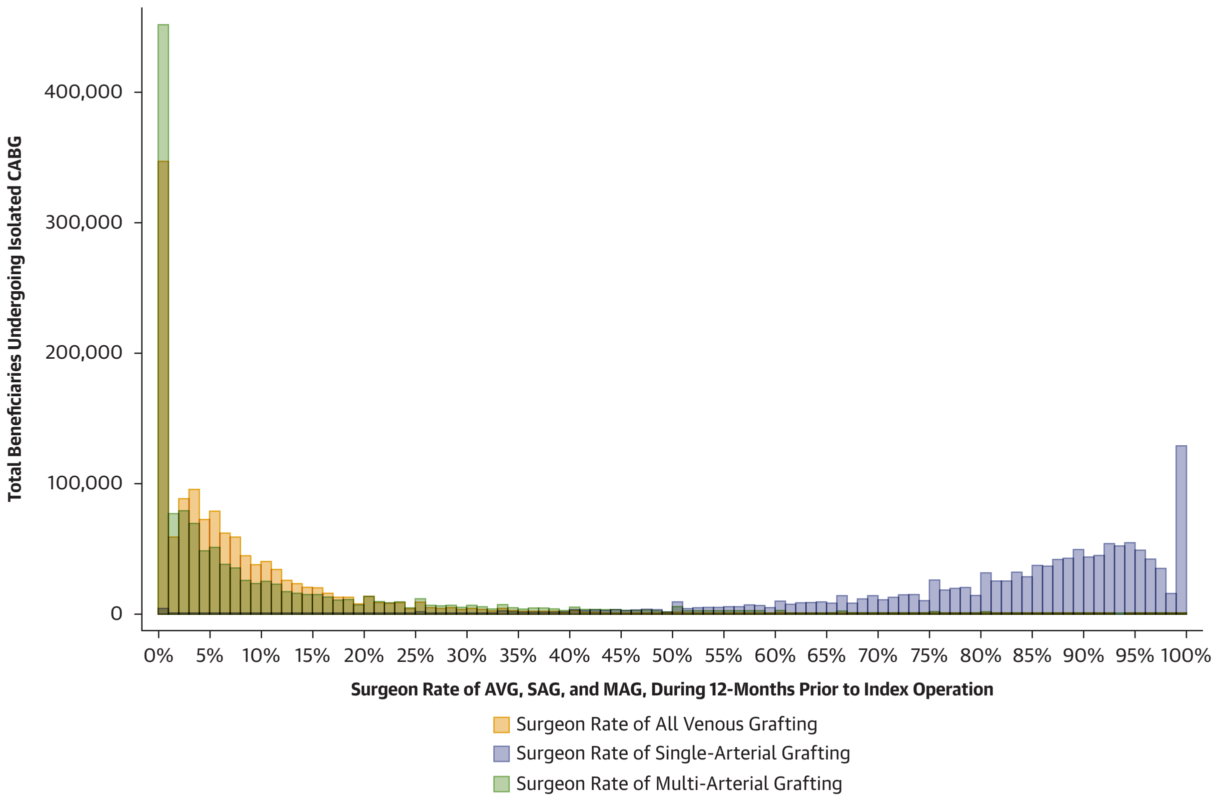
to address unmeasured confounding. Importantly, this null association was consistent across multiple sensitivity analyses (Table 2).

The contrasting results of the non-IV and IV models suggest that unmeasured confounding in the non-IV analysis may have led to an overestimation of the survival benefit attributable to MAG in conventional retrospective analyses. The null findings of our IV analysis are directionally consistent with the ART trial,[18] the only completed randomized study sufficiently powered to evaluate the long-term survival impact of MAG. However, direct comparison between our analysis and the ART trial is limited by important differences in study design and exposure definition. The ART trial specifically compared BIMA grafting with SIMA grafting, whereas our claims-based MAG classification encompasses heterogeneous MAG strategies, including BIMA, SIMA plus radial artery grafting, or combinations thereof. The ART trial also has recognized limitations, including substantial treatment crossover, concerns regarding right internal mammary artery harvesting, and enrollment of older patients less likely to derive long-term benefit. The ROMA trial, which completed enrollment in 2023, was designed to address several of these limitations by implementing an open MAG treatment strategy (allowing either BIMA or SIMA plus radial artery grafting), enrolling younger patients, and achieving lower crossover rates.[44] The awaited long-term outcomes from ROMA will provide important additional randomized evidence to further contextualize the findings of the present study.

STUDY LIMITATIONS. The current study is subject to several limitations. Although we implemented surgeon MAG rate as an IV to address limitations intrinsic to traditional retrospective analyses, our study remains retrospective and observational. Certain clinical details that influence conduit selection are not adequately captured in Medicare administrative claims. Nonetheless, we believe that risk adjustment using measured covariates (eg, pre-existing diagnoses of heart failure and myocardial infarction, total graft counts, etc), combined with our IV approach, helps to mitigate potential sources of bias.

Similarly, eligibility for MAG cannot be determined from administrative claims data. Consequently, some patients included in this analysis may

FIGURE 4 Surgeon Conduit Strategy Preferences



Histogram depicting surgeon rates of AVG, SAG, and MAG during the 12 months before the index operation for each Medicare beneficiary undergoing CABG. Abbreviations as in Figures 1 and 2.

not have been clearly eligible for both approaches. The IV estimate applies specifically to a latent subpopulation often referred to as "compliers" (patients whose treatment assignment is influenced by surgeon preference rather than clinical necessity). This population cannot be clinically identified or characterized in our data. Therefore, translation of these findings to specific patient groups, particularly those who may be optimal candidates for MAG based on favorable anatomy and conduit availability, should be made with caution. Our findings speak most directly to patients on the margin of receiving MAG vs SAG for whom surgeon preference plays a meaningful role in treatment selection.

Our analyses did not explicitly account for clustering at the hospital or surgeon level, which could result in underestimated standard errors if substantial within-provider correlation exists. However, the IV approach partially addresses surgeon-level confounding by design through between-surgeon variation in MAG use, and our models additionally include hospital and surgeon annual volume to capture important provider-level variation.

Procedural mischaracterization is a well-recognized concern in analyses using administrative claims data. To mitigate this risk, we excluded cases with discordant ICD and CPT codes regarding the presence of concomitant cardiac surgery.[23] We then used surgeon-billed CPT codes to determine the number and type of arterial and venous grafts (among other operative details).[26] CPT codes have been demonstrated to perform comparably with STS database documentation in capturing the number and type of bypass conduits described in operative dictations in a study conducted among several hospitals within a single health care system.[27] Further supporting the validity of this approach, the temporal trends in MAG use observed in our study closely mirror those reported in the STS database.[17]

The validity of IV analyses depends on assumptions that are largely untestable. Regarding the exclusion restriction, surgeon MAG rate may be

TABLE 2 Estimated Coefficients (95% CIs) for the Association of MAG (Compared to SAG) With Survival After CABG Using Differing Statistical Techniques and Definitions of the IV

	Non-IV-Model	IV-Model (IV: MAG rate prior 12 mo)	IV-Model (IV: MAG rate prior 36 mo)	IV-Model (IV: MAG rate prior 60 mo)	IV-Model (IV: MAG rate 1999-2019)
Beneficiaries with ≥10 years of follow-up undergoing CABG 2001-2009 (n = 834,724)[a]					
Logistic regression	-0.0948 (-0.110 to -0.0797)	0.0228 (-0.00743 to 0.0530)	0.0208 (-0.0106 to 0.0523)	0.0127 (-0.0199 to 0.0453)	0.00758 (-0.0246 to 0.0397)
Linear regression (2SLS)	-0.0200 (0.0230 to -0.0169)	0.00396 (-0.00236 to 0.0103)	0.00354 (-0.00308 to 0.0102)	0.00170 (-0.00516 to 0.00857)	0.000710 (-0.00608 to 0.00751)
Cox regression (2SRI)	-0.0655 (-0.0758 to -0.0551)	0.0117 (-0.00872 to 0.0322)	0.00820 (-0.0131 to 0.0295)	0.00224 (-0.0199 to 0.0244)	0.00192 (-0.0199 to 0.0237)
Flexible parametric (2SRI, without TDE)	-0.0657 (-0.763 to -0.0550)	0.0117 (-0.00875 to 0.0322)	-0.00818 (-0.0132 to 0.0295)	0.00218 (-0.0199 to 0.0243)	0.00193 (-0.0199 to 0.0237)
Flexible parametric (2SRI, with TDE)	[b]	[b]	[b]	[b]	[b]
Strength of IV: F-statistic (2SLS)	[c]	250,072	222,360	202,968	207,923
Strength of IV: LRT (2SRI)	[c]	124,664	115,000	107,414	113,305
Beneficiaries With any duration of follow-up undergoing CABG 2001-2019 (n = 1,275,606)[d]					
Logistic Regression	-0.103 (-0.117 to -0.0899)	0.000400 (-0.0267 to 0.0274)	-0.00330 (-0.0312 to 0.0246)	-0.00970 (-0.0385 to 0.0191)	-0.0201 (-0.0491 to 0.009)
Linear regression (2SLS)	-0.0174 (0.0198 to -0.0150)	-0.000130 (-0.00525 to 0.00498)	-0.00068 (-0.00599 to 0.00463)	-0.00200 (-0.0075 to 0.00351)	-0.00418 (-0.00976 to 0.00139)
Cox regression (2SRI)	-0.0707 (-0.0786 to -0.0628)	-0.00552 (-0.0212 to 0.0102)	-0.00972 (-0.0259 to 0.00648)	-0.0146 (-0.0313 to 0.00220)	-0.0138 (-0.0306 to 0.00306)
Flexible parametric (2SRI, without TDE)	-0.0707 (-0.0786 to -0.0628)	-0.00568 (-0.0214 to 0.0100)	-0.00988 (-0.0261 to 0.00633)	-0.0147 (-0.0315 to 0.00203)	-0.0139 (-0.0307 to 0.00292)
Flexible parametric (2SRI, with TDE)	[b]	[b]	[b]	[b]	[b]
Strength of IV: F-statistic (2SLS)	[c]	368,158	334,369	305,660	295,917
Strength of IV: LRT (2SRI)	[c]	176,733	166,008	155,404	153,252

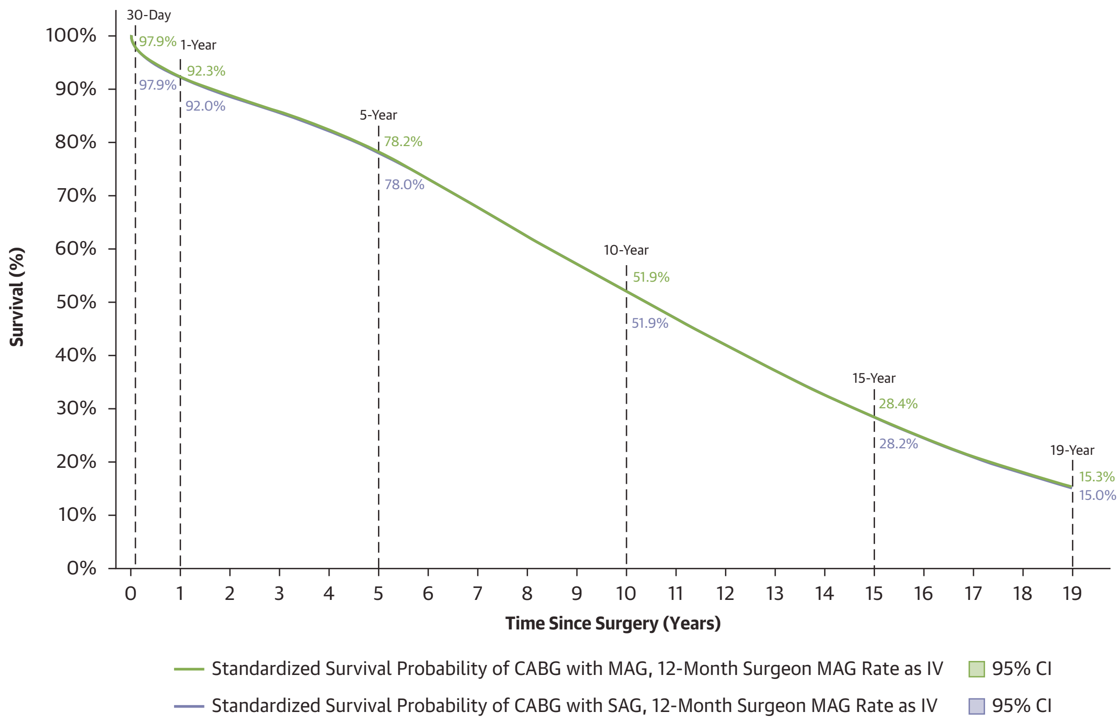
Values are estimated coefficients (95% CIs) or n. Statistical techniques included logistic, linear (2-stage least square [2SLS]), Cox (2-stage residual inclusion [2SRI]), and flexible parametric (2SRI without and with time-dependent effects) regressions used as sensitivity analyses. Coefficients are shown for both non-IV (column 2) and IV models (columns 3-6) using differing surgeon MAG rates (preceding 12 months, preceding 36 months, preceding 60 months, and overall [1999-2019]) as the instrument. For logistic and linear (2SLS) regressions, the dichotomous outcome was "survival status at 10 years." For Cox and flexible parametric regressions, the outcome was time-to-event using all available survival data. Finally, first-stage model comparisons with and without the IV (as measured by the F-statistic and likelihood-ratio test statistics for the 2SLS and 2SRI models, respectively) are reported in the bottom 2 rows of the table. These sensitivity analyses were completed in 2 study cohorts: (top half of table) beneficiaries undergoing CABG from 2001 to 2009 (thus, ensuring at least 10 years of survival data for each subject) and (bottom half of table) beneficiaries undergoing CABG from 2001 to 2019 (including beneficiaries with follow-up of <10 years). [a]Outcomes variables: 10-year survival (logistic, linear models), time to death at 10-years (Cox, flexible parametric models). [b]Coefficients omitted because the treatment effect is determined by the combination of multiple main effect and interaction coefficients. [c]Non-IV model, thus no F-statistic or LRT comparison performed. [d]Outcome variables: 10-year survival (logistic, linear models), time to death at 19-years (Cox, flexible parametric models).

2SLS = 2-stage least square (control function) IV analysis, used in linear regression model; 2SRI = 2-stage residual inclusion regression IV analysis, used in Cox and flexible parametric models; IV = instrumental variable; LRT = likelihood-ratio test statistic comparing IV model against non-IV model; TDE = time-dependent effects for the exposure variable (MAG vs SAG), modeled using restricted cubic splines with 3 degrees of freedom; other abbreviations as in Table 1.

correlated with other factors that independently affect survival, such as referral patterns, program-level practices, or surgical expertise. If surgeon MAG rate is associated with these or other unmeasured factors that independently influence survival, the IV estimate may be biased. Although we cannot definitively test these assumptions, we note that measurable patient characteristics were generally balanced across surgeon MAG rate quartiles after adjustment (Supplemental Table 6), and our results were consistent across multiple IV specifications and statistical approaches (**Table 2**). In this context, ongoing debate persists regarding the extent to which correlations among surgeon volume, practice patterns, and case mix may influence IV-based estimates and whether these estimates are more or less reflective of the true treatment effect.[45]

Ultimately, the ability of surgeon preference IV analyses to approximate treatment effects can only be validated by comparing the results of these analytic approaches with randomized controlled trials. We have applied surgeon preference IV techniques to analyze other topics in cardiac surgery and derived results concordant with randomized studies. For example, in an analysis comparing mitral valve repair vs replacement during CABG for ischemic mitral regurgitation,[46] the IV results (as compared with non-IV results) more closely approximated the 2-year results of a randomized controlled trial[47] conducted

FIGURE 5 Long-Term Survival After CABG (IV-Model)



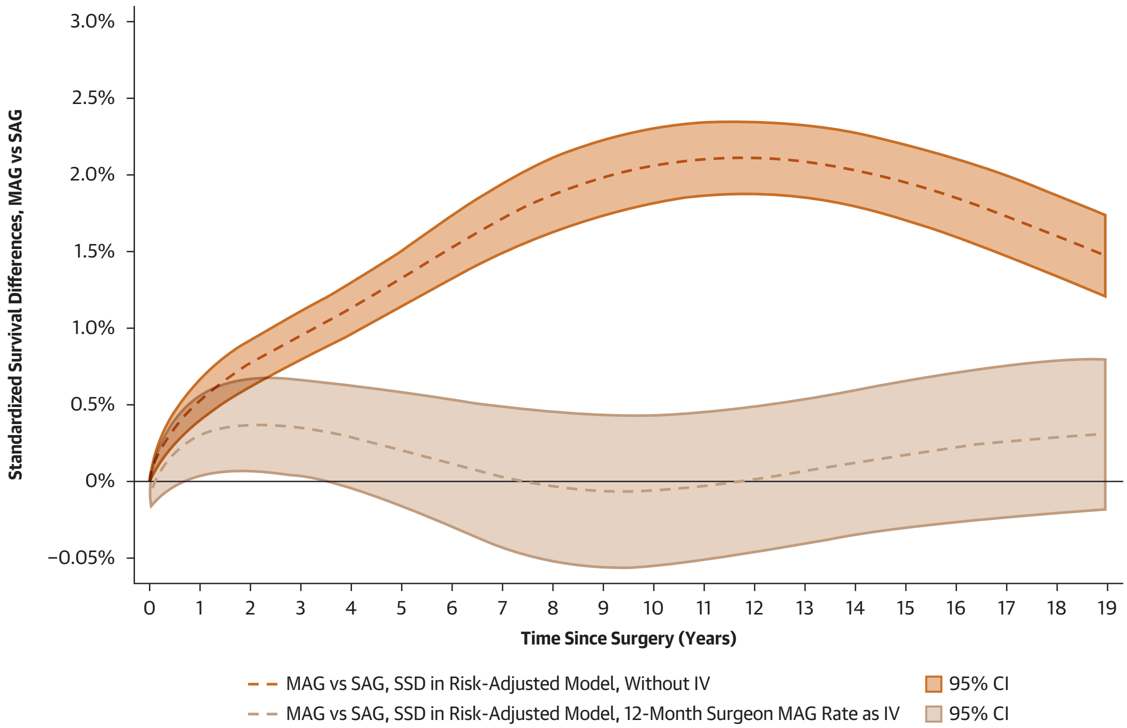
Standardized survival probabilities after CABG in the IV model (a model including surgeon MAG rate as an instrumental variable); stratified by whether the beneficiary received SAG (purple) or MAG (green). Abbreviations as in Figures 1 to 3.

in a similar patient population. In another study,[48] the time-dependent treatment effect of surgical ablation for pre-existing atrial fibrillation during CABG estimated using an IV analytic approach closely mirrored the survival benefit observed among heart failure patients undergoing catheter ablation in the CASTLE-AF randomized trial.[49] In the present study, we reach a similar conclusion: The treatment effect estimates generated by the IV model more closely align with available randomized data than do results from the non-IV model.

We did not evaluate other major adverse cardiovascular events such as repeat revascularization or myocardial infarction because we have not yet rigorously assessed the validity and consistency of these endpoints within our dataset. However, the IV analytic methodology demonstrated in the current analysis can be extended to these endpoints in future work. It is possible that although graft patency may be higher among MAG recipients,[15,16] advances in percutaneous coronary intervention may mitigate the long-term survival impact of graft failure to non-left anterior descending targets, thereby contributing to the neutral survival results observed in our IV analysis. Interestingly, the time-dependent nature of the treatment effect noted in the present IV analysis suggests that any survival benefit of arterial grafting may be concentrated in the early to midterm period (approximately 1-7 years postoperatively), an observation requiring validation in future studies.

Prior retrospective analyses suggest that the survival benefit of MAG over SAG may be more pronounced in younger patients.[17] Given the mean age of 72 years in our cohort, our findings may not be generalizable to a younger population. Replicating our study design in a younger cohort, for example, using STS data linked to sources of long-term survival, would provide valuable additional insight.

FIGURE 6 **Standardized Survival Difference Between MAG and SAG**



Nineteen-year standardized survival differences after CABG between SAG and MAG recipients in the non-IV model (copper) and IV model (tan). Surgeon MAG rate in the 12 months preceding index CABG was used as the IV in the primary analysis. Abbreviations as in Figures 1 to 3.

CONCLUSIONS

The results of this study highlight important discrepancies between 2 analytic approaches to assess the association of MAG vs SAG with survival after CABG. Our non-IV model, consistent with prior conventional retrospective observational studies, suggested a modest survival benefit of MAG over SAG. However, this benefit was not observed in the IV model. Notably, results of the IV model align with the 10-year survival results of the ART trial, which currently represents the best available randomized evidence.[18] These findings underscore the need for cautious interpretation of observational studies and emphasize the importance of eagerly anticipated high-quality randomized controlled trials (such as ROMA and ROMA-W)[46] to more definitively characterize the treatment effect of MAG vs SAG in CABG.

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KEY WORDS coronary artery bypass grafting, instrumental variable analysis, multiarterial grafting, surgeon preference, unmeasured confounding

APPENDIX For an expanded Methods section and supplemental tables and a figure, please see the online version of this paper.